

A Current mortality risk

ARI registry, Ecuador, 2015–2023

576,195 admissions; 41,048 in-hospital deaths
33 patient and historical-context features

Historical facility service profiles

9,463,108 non-ARI discharges; preceding 12 months
7 service statistics + 11 diagnostic proportions

Base learning and stratified calibration

LightGBM + XGBoost; nested predictions

Stacking → residual correction → marginal projection

$$p_i^H = \sigma(\beta_0 + \beta_1 z_i^B + \sum_k \gamma_k g_k(i))$$

$$\text{Fitting target } \sum_i g_k(i)(y_i - p_i^H) \approx 0$$

Monthly probability maintenance

$$\text{logit}(p_{i,t}) = \phi_i^\top \theta_t$$

$$\phi_i = [1, z_i^H, v_i, c_i]^\top, \quad \theta_t = \theta_{t-1} + \varepsilon_t$$

$\varepsilon_t \sim N(0, 0.1I)$; predict before updating

Predictive evaluation

Cross-validation AUROC 0.9037 (EW 0.9028)

Mean marginal $|\log(O/E)|$: 0.0219 → 0.0054

Temporal ICI: 0.0196 static → 0.0052 updated

EW with the same update has similar performance

B Independent monitoring references

Regional and facility comparisons

Independent clinical case-mix LightGBM

$$E_g = \sum_{i \in g} q_i, \quad O_g = \sum_{i \in g} y_i$$

$$\text{SMR}_g = O_g / E_g; \text{ Gamma-Poisson shrinkage}$$

Provincial ratios 0.61–1.56; 10 above reference

Non-COVID historical outcomes

Fix the clinical model fitted in 2015–2019

2020–2021: $O = 3,485$, $E = 2,108$

$O/E = 1.65$ (95% interval 1.60–1.71)

2022–2023: $O/E = 0.97$

Paediatric admission demand

Pre-pandemic seasonal Poisson count reference

Population offset; trend sensitivity analysis

2023 admissions: $O/E = 1.32$ (95% PI 1.23–1.43)

Deaths: 1.00 (0.84–1.19); trend-adjusted admissions:
1.11

Interpretation for further investigation

Regional departure, historical change and seasonal demand

Prioritise record review and local service assessment

Prediction and monitoring share registry records but use different expected values. q_i is an independent reference probability, not the Hi-CARE output. Marginal constraints apply to fitting data. Province holdout separately evaluates a profile ensemble without the full second layer. PI, prediction interval.